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A PROJECT REPORT

ON

"Aletheia: A Multi-Engine Retrieval-Augmented Architecture for Real-Time Hallucination Detection and Automated Fact Verification in Large Language Models"

A project report submitted in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By
Suparshw Rinayate (25070521231)
Nihar Ilame (25070521233)
Pranjal Choudhary (25070521234)

UNDER THE GUIDANCE OF

Dr. Pratik Agrawal

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Chapter 1: Introduction

1.1 Introduction

Large Language Models (LLMs) such as GPT-4, LLaMA-3 and Claude 3 generate text token by token without any built-in check against real-world facts, so a fluent answer can still be false. This is called hallucination, and it is most dangerous in medicine, law and scientific research. The paper groups hallucinations into three types: **intrinsic** (contradicts the source or ground truth), **extrinsic** (cannot be verified or refuted from the context) and **factual drift** (the answer departs from earlier facts over a multi-turn conversation).

1.2 Problem Statement

Most fact-checking and Retrieval-Augmented Generation (RAG) systems rely on one knowledge source, usually Wikipedia or a single web search. If that retriever returns incomplete or biased data, the error flows into the verdict. Judging a long answer as a whole is also unreliable, because it can mix true and false statements. The problem is to verify LLM output claim by claim, from several independent sources, with an audit trail and a correction.

1.3 Scope, Hypothesis and Objectives

Aletheia (v1.0_FACTUALITY_CORE) audits any LLM response in real time and is evaluated on TruthfulQA, FEVER and HALU-Eval. The hypothesis is that cross-verifying each atomic claim against heterogeneous sources detects hallucinations better than a single source, at low latency because retrieval runs in parallel. The objectives of this report are:

- To document the four-stage pipeline for hallucination detection and revision.
- To analyse the reported gains in precision and false-positive reduction.
- To judge the strength of the evidence and list gaps and improvements.

Chapter 2: Literature Review

RAG grounds a model in retrieved documents [1], but a single retriever is a single point of failure. Hallucination surveys show the problem across summarisation, dialogue and question answering [2]. FActScore showed that splitting long text into atomic claims greatly improves fact-checking precision [3]. FEVER [4] and TruthfulQA [5] are standard benchmarks. **Research gap:** existing checkers use one source. Aletheia combines Wikipedia, web search, academic repositories (ArXiv, Semantic Scholar, OpenAlex) and a private local vector store, and adds an auditable ledger.

Chapter 3: Software Requirements Specification

The paper does not state hardware or cost figures; it notes that parallel API calls raise network overhead and query cost. The software stack is in Table 3.1.

Table 3.1: Technology stack of the Aletheia platform

Layer	Technology
Backend	FastAPI (Python 3.11), PyTorch, HuggingFace Transformers, LangChain, asyncio
Embeddings	OpenAI text-embedding-3-small or bge-large-en (local vector store)
Retrieval APIs	Wikipedia; Google/DuckDuckGo; ArXiv, Semantic Scholar, OpenAlex
Frontend	Next.js, Tailwind CSS, Lucide icons, dark-mode interface (port 8000)

Functional requirements: accept a question and answer; extract atomic claims; query four engines concurrently; classify each claim as VERIFIED, HALLUCINATION or UNVERIFIED; score overall factuality; revise contradicted claims and export an evidence ledger. **Non-functional requirements:** latency under 1.8 s for a 200-word response, high precision with few false positives, full auditability, and controlled API cost.

Chapter 4: Methodology and Implementation

Aletheia separates generation from verification and runs four stages on each response (Figure 4.1).

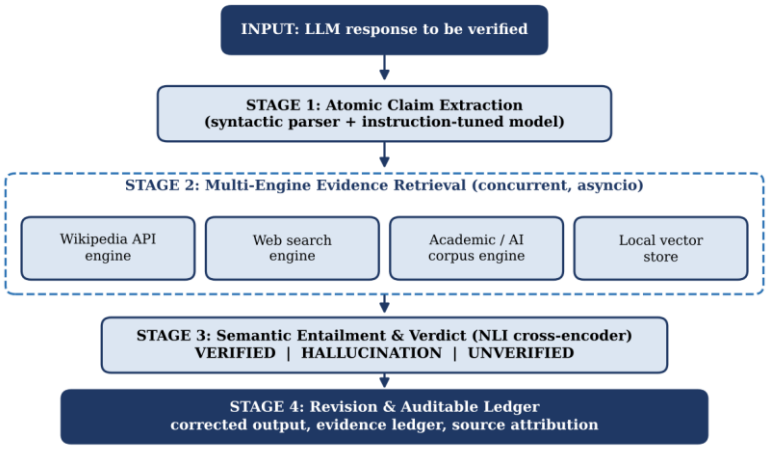


Figure 4.1: Four-stage Aletheia verification pipeline

Stage 1, claim extraction: a rule-based parser and an instruction-tuned model split the answer into atomic claims, each with one subject-predicate-object relation and no unresolved pronouns. **Stage 2, retrieval:** four engines run concurrently with asyncio: Wikipedia, web search, academic corpus, and a local vector store using cosine similarity. **Stage 3, verdict:** an NLI cross-encoder scores each claim-passage pair as entailment (VERIFIED), contradiction (HALLUCINATION) or neutral (UNVERIFIED), and a weighted factuality score is computed. **Stage 4, revision:** contradicted claims are rewritten

from the retrieved evidence, and every step is saved to an auditable ledger with source attribution.

Chapter 5: Testing and Validation

The paper evaluates on TruthfulQA (misconceptions), FEVER (claim verification) and HALU-Eval (QA, dialogue, summarisation) against three baselines: vanilla GPT-3.5-Turbo, Wikipedia-only RAG and web-search-only RAG. It reports precision, false-positive reduction and latency, but not recall, sample sizes or the NLI model used.

Chapter 6: Results and Discussion

Table 6.1: Precision of Aletheia and baseline systems

Model / System	**Precision (%)**
Vanilla GPT-3.5-Turbo	58.4
Single-source (Wikipedia RAG)	76.4
Single-source (Web Search RAG)	81.2
Aletheia (Multi-Engine Core)	92.8

Aletheia is 11.6 points above the best baseline. Combining web, Wikipedia and academic sources cut false hallucination flags by 34.5% versus Wikipedia-only checking, and latency stays under 1.8 s for a 200-word response. **Strengths:** claim-level checking, four complementary sources, an auditable ledger, and a separate "unverified" verdict. **Authors' limits:** stale web evidence, API cost and rate limits, and metaphors that confuse the claim extractor. **Gaps in the evidence:**

- Only precision is reported. Recall, F1 and accuracy are missing, although the abstract claims a recall gain.
- The metric is undefined, and a vanilla LLM is not a like-for-like baseline. Comparison with FActScore or SelfCheckGPT would be fairer.
- There are no engine ablations, per-benchmark results or confidence intervals, and the NLI, extraction and revision models are unnamed.
- The conclusion says hallucination risks are eliminated, which 92.8% precision does not support.

Chapter 7: Conclusion and Future Work

Aletheia is a well-structured approach to LLM fact checking: atomic claims, four parallel evidence sources, NLI verdicts and an auditable revision step address the single-source weakness of standard RAG. The 92.8% precision is promising, but a fuller evaluation and named models are needed to trust it. **Future work:** multi-modal verification of vision-

language outputs, Neo4j knowledge-graph integration, and local fine-tuned NLI models that run offline, plus recall/F1 reporting and engine ablations.

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